

Route Blocking for Reducing Forest Fire Spread

COL761 Assignment 2 - Q2

1 Introduction

We study the problem of minimizing wildfire spread in a forest modeled as a directed probabilistic graph. Each edge represents a possible route of fire propagation with an associated probability. Given a limited budget k , the objective is to block k edges such that the expected number of burned nodes is minimized.

2 Task 1: Optimization Formulation

2.1 Decision Variables

Let $G = (V, E, p)$ be a directed graph. For each edge $e \in E$, define a binary decision variable:

$$x_e = \begin{cases} 1 & \text{if edge } e \text{ is blocked,} \\ 0 & \text{otherwise.} \end{cases}$$

Let $R = \{e \in E \mid x_e = 1\}$ denote the set of blocked edges.

2.2 Objective Function

Let $\sigma(R)$ denote the expected number of burned regions after removing a set of edges $R \subseteq E$. The objective is:

$$\min_{R \subseteq E} \sigma(R) = \mathbb{E}[|A_\infty|]$$

where the A_∞ denotes the set of burned nodes at termination and expectation is taken over the stochastic fire propagation process.

2.3 Constraints

The number of blocked edges is limited by the budget k :

$$|R| = k$$

2.4 Final Formulation

Thus, the problem can be written as:

$$\min_{R \subseteq E} \sigma(R) \quad \text{s.t.} \quad |R| = k$$

where $\sigma(R) = \mathbb{E}[|A_\infty|]$ denotes the expected number of burned nodes at termination.

The route blocking problem is a combinatorial optimization problem, where the goal is to select a subset $R \subseteq E$ of size k that minimizes $\sigma(R)$.

3 Task 2: NP-Hardness

We prove that the route blocking problem is NP-hard via a reduction from the Influence Minimization problem under the Independent Cascade (IC) model, which is known to be NP-hard.

3.1 Reduction

In the Influence Minimization problem, we are given a directed graph $G = (V, E, p)$ and a seed set $A_0 \subseteq V$. The goal is to remove a set of k edges such that the expected spread of influence is minimized.

Given an instance of the Influence Minimization problem, we construct an identical instance of the route blocking problem:

- The graph $G = (V, E, p)$ remains unchanged.
- The seed set A_0 is the same.
- The propagation model is identical (Independent Cascade model).
- The budget k remains the same.

3.2 Correctness

Under this construction, the expected number of influenced nodes in the Influence Minimization problem is exactly equal to the expected number of burned nodes in the route blocking problem, i.e., $\sigma(R)$.

Thus, selecting k edges to remove in order to minimize influence spread is identical to selecting k edges to block in order to minimize fire spread.

Therefore, any optimal solution to the route blocking problem yields an optimal solution to the Influence Minimization problem.

3.3 Conclusion

Since the reduction is clearly computable in polynomial time and Influence Minimization is NP-hard, it follows that the route blocking problem is NP-hard.

4 Task 3: Structural Properties

Let $\sigma(R)$ denote expected spread after removing edges R .

4.1 Monotonicity

Claim: $\sigma(R)$ is monotone non-increasing.

Proof: Let $R_1 \subseteq R_2$. Then:

$$E \setminus R_2 \subseteq E \setminus R_1$$

Thus, the graph corresponding to R_2 has fewer edges than that of R_1 , implying fewer possible propagation paths.

For any realization of the stochastic process (i.e., a fixed set of active edges), the set of nodes reachable from A_0 under R_2 is a subset of that under R_1 . Hence:

$$|A_\infty(R_2)| \leq |A_\infty(R_1)|$$

Taking expectation over all realizations:

$$\sigma(R_2) \leq \sigma(R_1)$$

Thus, $\sigma(R)$ is monotone non-increasing.

4.2 Submodularity

Claim: $\sigma(R)$ is submodular.

Proof: Consider a random realization of the graph, where each edge is independently retained with its probability. In any fixed realization, the fire spreads deterministically along the active edges.

Let $f_R(\omega)$ denote the number of nodes reachable from A_0 in realization ω after removing edges R .

For a fixed realization ω , $f_R(\omega)$ is a deterministic reachability function. Removing edges can only reduce reachability, and it is well known that reachability is a submodular function with respect to edge deletions. That is, for $A \subseteq B \subseteq E$ and $e \notin B$:

$$f_A(\omega) - f_{A \cup \{e\}}(\omega) \geq f_B(\omega) - f_{B \cup \{e\}}(\omega)$$

Taking expectation over all realizations ω , we obtain:

$$\sigma(A) - \sigma(A \cup \{e\}) \geq \sigma(B) - \sigma(B \cup \{e\})$$

Thus, $\sigma(R)$ satisfies the diminishing returns property and is submodular.

5 Task 4: Algorithm Design

5.1 Overview

We propose a hybrid heuristic combining:

- Cluster-based structural edge removal
- Seed-based greedy removal

We evaluate both strategies and select the better one.

5.2 Algorithm

Algorithm 1 Hybrid Route Blocking Algorithm

- 1: Construct graph G
- 2: Compute initial spread $\sigma(\emptyset)$

Approach 1: Cluster-Based Edge Removal

- 3: Generate clusters using MCL
- 4: Identify bridge edges between clusters
- 5: Compute Personalized PageRank (PPR) from seeds
- 6: **for** each bridge edge (u, v) **do**
- 7: Compute score: $\frac{\text{PPR}(u) \cdot p(u, v)}{\text{PPR}(v)}$
- 8: **end for**
- 9: Select top- k edges R_1

Approach 2: Seed-Based Greedy Removal

- 10: Select top- k edges from seed neighbors based on probability R_2

Selection via Simulation

- 11: Estimate $\sigma(R_1)$ and $\sigma(R_2)$ using simulation
 - 12: **if** $\sigma(R_1) < \sigma(R_2)$ **then**
 - 13: $R \leftarrow R_1$
 - 14: **else**
 - 15: $R \leftarrow R_2$
 - 16: **end if**
 - 17: **return** R
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5.3 Complexity Analysis

Let:

- $n = |V|$, $m = |E|$
- r = number of simulations
- Graph construction: $O(m)$
- Markov clustering (MCL): $O(I \cdot m)$, where I is the number of MCL iterations
- Bridge edge identification: $O(m)$
- Personalized PageRank: $O(T \cdot m)$, where T is the number of PageRank iterations
- Edge scoring and sorting: $O(m \log m)$
- Simulation: $O(r \cdot (n + m))$

Thus, the total complexity is:

$$O(I \cdot m + T \cdot m + m \log m + r(n + m))$$

In practice, the simulation term $O(r(n + m))$ dominates the runtime.

5.4 Approximation Guarantees

The objective function $\sigma(R)$ is monotone and submodular. Therefore, a greedy algorithm provides a $(1 - 1/e)$ approximation guarantee.

However, our method:

- Uses heuristic scoring instead of exact marginal gains
- Does not strictly follow greedy selection

Thus, **no formal approximation guarantee** can be claimed.

6 Task 5: Evaluation

Let:

$$\rho(R) = \frac{\sigma(\emptyset) - \sigma(R)}{\sigma(\emptyset)}$$

- Dataset 1: $\rho(R) = \text{---}$
- Dataset 2: $\rho(R) = \text{---}$

7 Conclusion

We proposed a hybrid approach combining clustering-based structural insights and seed-centric heuristics. By evaluating both strategies via simulation, the method adapts to different graph structures and achieves effective reduction in expected fire spread.